

Mathematical and Logical Foundations of Computer Science

Lecture 7 - Propositional Logic (Semantics)

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(some slides were adapted from Rajesh Chitnis' slides)

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Where are we?

- ▶ Symbolic logic
- ▶ **Propositional logic**
- ▶ Predicate logic

Today

- ▶ semantics of propositional logic
- ▶ satisfiability & validity
- ▶ truth tables
- ▶ soundness & completeness

Further reading:

- ▶ Chapter 6 of
http://leanprover.github.io/logic_and_proof/

Recap: Propositional logic syntax

Syntax:

$$P ::= a \mid P \wedge P \mid P \vee P \mid P \rightarrow P \mid \neg P$$

Two special atoms:

- ▶ $\top$ which stands for True
- ▶ $\perp$ which stands for False

We also introduced four connectives:

- ▶ $P \wedge Q$: we have a proof of both P and Q
- ▶ $P \vee Q$: we have a proof of at least one of P and Q
- ▶ $P \rightarrow Q$: if we have a proof of P then we have a proof of Q
- ▶ $\neg P$: stands for $P \rightarrow \perp$

Syntax vs. Semantics

Syntax

- ▶ Rules for allowable formulas in the language
- ▶ Syntax for propositional logic:

$$P ::= a \mid P \wedge P \mid P \vee P \mid P \rightarrow P \mid \neg P$$

Semantics

- ▶ Assigning meaning/interpretations with formulas
- ▶ Semantics for propositional logic: This lecture!

Syntax and Semantics for the English language?

- ▶ Syntax: alphabet and grammar
- ▶ Semantics: meanings for words

Semantics for Propositional Logic

Semantics assigns **meanings/interpretations** with **formulas**

The basic notion we use is **“truth value”**

The two standard truth values are “true” and “false”

We use the symbols **T** and **F** respectively

This is a **classical** notion of truth

- ▶ i.e., interpretation of each proposition is either true or false
- ▶ **Excluded Middle**: for each A we have $A \vee \neg A$
- ▶ Here it means for each A , we have that A is either true or false.

WARNING: This is just one possible way to assign meanings!

Semantics for Propositional Logic (continued)

Truth assignment

- ▶ Function assigning a truth value for each atomic proposition
- ▶ E.g., given 2 atomic propositions p, q , if the formula is $p \vee q$
- ▶ then one truth assignment ϕ is $\phi(p) = \mathbf{T}$ and $\phi(q) = \mathbf{F}$
- ▶ Also called an “interpretation” or a “valuation”

How many truth valuations do we need to consider for $p \vee q$?

- ▶ $2^2 = 4$
- ▶ $\phi(p) = \mathbf{T}, \phi(q) = \mathbf{T}$ and $\phi(p) = \mathbf{T}, \phi(q) = \mathbf{F}$ and
 $\phi(p) = \mathbf{F}, \phi(q) = \mathbf{T}$ and $\phi(p) = \mathbf{F}, \phi(q) = \mathbf{F}$

Conventions:

- ▶ The atoms $\top, \perp$ have the interpretations $\mathbf{T}, \mathbf{F}$ respectively
- ▶ $\phi(\top) = \mathbf{T}$ and $\phi(\perp) = \mathbf{F}$

Semantics of logical connectives

How to extend the notion of semantics to **compound formulas**?

Define semantics for the four logical connectives: $\vee, \wedge, \rightarrow, \neg$

This is done **recursively bottom-up** over the structure of propositions.

For example given a conjunction $A \wedge B$, we first have to evaluate the truth-values of A and B to compute the truth-value of $A \wedge B$.

I.e., $\phi(A \wedge B) = \mathbf{T}$ iff both $\phi(A) = \mathbf{T}$ and $\phi(B) = \mathbf{T}$.

Semantics of logical connectives

The **extended valuation function** is recursively defined as follows:

- ▶ $\phi(\top) = \mathbf{T}$
- ▶ $\phi(\perp) = \mathbf{F}$
- ▶ $\phi(A \vee B) = \mathbf{T}$ iff either $\phi(A) = \mathbf{T}$ or $\phi(B) = \mathbf{T}$
- ▶ $\phi(A \wedge B) = \mathbf{T}$ iff both $\phi(A) = \mathbf{T}$ and $\phi(B) = \mathbf{T}$
- ▶ $\phi(A \rightarrow B) = \mathbf{T}$ iff $\phi(B) = \mathbf{T}$ whenever $\phi(A) = \mathbf{T}$
- ▶ $\phi(\neg A) = \mathbf{T}$ iff $\phi(A) = \mathbf{F}$

Semantics of logical connectives

What is $\phi(2 > 1 \wedge 1 > 0)$? (inequalities are atomic propositions)

$\phi(2 > 1 \wedge 1 > 0) = \mathbf{T}$ because $\phi(2 > 1) = \mathbf{T}$ and $\phi(1 > 0) = \mathbf{T}$

What is $\phi(2 > 1 \wedge 0 > 1)$?

$\phi(2 > 1 \wedge 0 > 1) = \mathbf{F}$ because $\phi(0 > 1) = \mathbf{F}$

What is $\phi(x > 1 \wedge 3 > x)$?

we don't know: it depends on $\phi(x > 1)$ and $\phi(3 > x)$

What is $\phi(x > 1 \vee 2 > x)$?

it depends on $\phi(x > 1)$ and $\phi(2 > x)$

$\phi(x > 1 \vee 2 > x) = \mathbf{T}$ for all combinations

only 2 possible combinations (the atoms are interdependent):

$\phi(x > 1) = \mathbf{T}, \phi(2 > x) = \mathbf{F}$ and $\phi(x > 1) = \mathbf{F}, \phi(2 > x) = \mathbf{T}$

Semantics of logical connectives

What is $\phi(2 > 0 \rightarrow 1 > 0)$? (inequalities are atomic propositions)

$\phi(2 > 0 \rightarrow 1 > 0) = \mathbf{T}$ because $\phi(1 > 0) = \mathbf{T}$

What is $\phi(0 > 2 \rightarrow 1 > 0)$?

still $\phi(0 > 2 \rightarrow 1 > 0) = \mathbf{T}$ because $\phi(1 > 0) = \mathbf{T}$

What is $\phi(2 > 0 \rightarrow 0 > 1)$?

$\phi(2 > 0 \rightarrow 0 > 1) = \mathbf{F}$ because $\phi(0 > 1) = \mathbf{F}$ while $\phi(2 > 0) = \mathbf{T}$

What is $\phi(0 > 2 \rightarrow 0 > 1)$?

$\phi(0 > 2 \rightarrow 0 > 1) = \mathbf{T}$ because $\phi(0 > 2) = \mathbf{F}$

What is $\phi(x > 2 \rightarrow x > 1)$? it depends on $\phi(x > 2)$ and $\phi(x > 1)$

$\phi(x > 2 \rightarrow x > 1) = \mathbf{T}$ for all possible combinations (the atoms are interdependent): $\phi(x > 2) = \mathbf{T}, \phi(x > 1) = \mathbf{T}$ and

$\phi(x > 2) = \mathbf{F}, \phi(x > 1) = \mathbf{T}$ and $\phi(x > 2) = \mathbf{F}, \phi(x > 1) = \mathbf{F}$

Satisfiability & Validity

The above technique allows answering the following question:

What is the truth value of a formula w.r.t. a given valuation of its atoms?

To analyze the meaning of a formula, we also want to analyze its truth value w.r.t. **all possible combinations** of assignments of truth values with its atoms.

Satisfaction & validity

- ▶ Given a valuation ϕ on all atomic propositions, we say that ϕ **satisfies** A if $\phi(A) = \mathbf{T}$.
- ▶ A is **satisfiable** if there exists a valuation ϕ on atomic propositions such that $\phi(A) = \mathbf{T}$.
- ▶ A is **valid** if $\phi(A) = \mathbf{T}$ for all possible valuations ϕ .

A method to check satisfiability and validity: **truth tables**

Truth tables

Semantics for “or”

$\phi(A \vee B) = \mathbf{T}$ iff either $\phi(A) = \mathbf{T}$ or $\phi(B) = \mathbf{T}$

Truth table for “or”

A	B	$A \vee B$
T	T	T
T	F	T
F	T	T
F	F	F

- ▶ One row for each valuation
- ▶ Last column has the truth value for the corresponding valuation

Truth tables

Semantics for “and”

$\phi(A \wedge B) = \mathbf{T}$ iff both $\phi(A) = \mathbf{T}$ and $\phi(B) = \mathbf{T}$

Truth table for “and”

A	B	$A \wedge B$
T	T	T
T	F	F
F	T	F
F	F	F

Truth tables

Semantics for “implies”

$\phi(A \rightarrow B) = \mathbf{T}$ iff $\phi(B) = \mathbf{T}$ whenever $\phi(A) = \mathbf{T}$

Truth table for “implies”

A	B	$A \rightarrow B$
T	T	T
T	F	F
F	T	T
F	F	T

Truth tables

Semantics for “not”

$$\phi(\neg A) = \mathbf{T} \text{ iff } \phi(A) = \mathbf{F}$$

Truth table for “not”

A	$\neg A$
T	F
F	T

Semantics for compound formulas

We can now construct a truth table for any propositional formula

- ▶ consider all possible truth assignments for the atoms
- ▶ then use truth tables for each connective recursively

What is the truth table for $(p \rightarrow q) \wedge \neg q$?

p	q	$p \rightarrow q$	$\neg q$	$(p \rightarrow q) \wedge \neg q$
T	T	T	F	F
T	F	F	T	F
F	T	T	F	F
F	F	T	T	T

- ▶ 2 atoms, and hence $2^2 = 4$ rows (one per interpretation)
- ▶ Use intermediate columns to evaluate sub-formulas
- ▶ 2 atoms and 3 connectives hence $2 + 3 = 5$ columns
- ▶ Rightmost column gives values of the formula

Satisfiability & validity

A formula is **satisfiable** iff there is a valuation that satisfies it
i.e., if there is a **T** in the rightmost column of its truth table

example: $p \wedge q$ because of the valuation $\phi(p) = \mathbf{T}, \phi(q) = \mathbf{T}$

A formula is **falsifiable** iff there is a valuation that makes it false
i.e., if there is a **F** in the rightmost column of its truth table

example: $p \wedge q$ because of the valuation $\phi(p) = \mathbf{F}, \phi(q) = \mathbf{T}$

A formula is **unsatisfiable** iff no valuation satisfies it
i.e., the cells of the rightmost column of its truth table all contain **F**

example: $p \wedge \neg p$ (contradiction)

A formula is **valid** iff every valuation satisfies it
i.e., the cells of the rightmost column of its truth table all contain **T**

example: $p \vee \neg p$ (tautology)

Validity of arguments using semantics

Validity of an argument

- ▶ **syntactically**: we can derive the conclusion from the premises
- ▶ **semantically**: the conclusion is true whenever the premises are

Formally, we write

$$P_1, \dots, P_n \models C$$

if the corresponding argument is **semantically valid**

i.e., every valuation that evaluates each of the premises $P_1, \dots, P_n$ to **T** also evaluates the conclusion C to **T**

Checking validity

- ▶ Already seen how to do this using “natural deduction”
- ▶ Truth tables is yet another way
- ▶ Bonus: yields counterexample if argument is invalid

Checking (semantic) validity

Is $P \rightarrow Q, \neg Q \models \neg P$ (semantically) valid?

P	Q	$P \rightarrow Q$	$\neg Q$	$\neg P$
T	T	T	F	F
T	F	F	T	F
F	T	T	F	T
F	F	T	T	T

Argument is valid: any row where conclusion is **F** then at least one of the premises is also **F**

Note that checking $P_1, \dots, P_n \models C$ is equivalent to checking the validity of $P_1 \rightarrow \dots P_n \rightarrow C$

i.e., that the cells of the rightmost column of the truth table for $P_1 \rightarrow \dots P_n \rightarrow C$ all contain **T**

Checking (semantic) validity

Is $\neg P \rightarrow \neg R, R \models \neg P$ (semantically) valid?

P	R	$\neg P$	$\neg R$	$\neg P \rightarrow \neg R$	R	$\neg P$
T	T	F	F	T	T	F
T	F	F	T	T	F	F
F	T	T	F	F	T	T
F	F	T	T	T	F	T

Argument is invalid

- ▶ Look at the first row
- ▶ Conclusion is **F**, but both premises are **T**
- ▶ Can we add a premise to make the argument valid?
 - ▶ Yes, we can add $\neg R$, which would be **F** in the first row

Proving anything using contradictions!

Is $P, \neg P \models C$ is (semantically) valid?

P	C	$\neg P$	C
T	T	F	T
T	F	F	F
F	T	T	T
F	F	T	F

Argument is (trivially) valid:

- ▶ Look at any row (we only have to look at rows where the conclusion is **F**)
- ▶ One of P and $\neg P$ is **F**

Truth Tables vs. Natural Deduction

Pros and cons of two ways of checking validity

Truth tables	Natural deduction
shows validity in a restricted setting (Boolean truth values)	checks validity in general setting (by an actual proof!)
simple, easy to automate	more difficult to automate
size of truth table is huge: exponential in number of atoms	typically scales better than brute force search
generates counterexamples if invalid	no easy way to check validity (other than actually proving)

Soundness & Completeness

Given a deduction system such as Natural deduction, a formula is said to be **provable** if there is a proof of it in that deduction system

- ▶ This is a **syntactic** notion
- ▶ it asserts the existence of a syntactic object: a proof
- ▶ typically written $\vdash A$

A formula A is **valid** if $\phi(A) = \mathbf{T}$ for all possible valuations ϕ

- ▶ it is a **semantic** notion
- ▶ it is checked w.r.t. valuations that give meaning to formulas

Soundness: a deduction system is sound w.r.t. a semantics if every provable formula is valid

- ▶ i.e., if $\vdash A$ then $\models A$

Completeness: a deduction system is complete w.r.t. a semantics if every valid formula is provable

- ▶ i.e., if $\models A$ then $\vdash A$

Soundness & Completeness

Classical Natural Deduction is

- ▶ **sound** and
- ▶ **complete**

w.r.t. the **truth table semantics**

Proving those properties is done within the **metatheory**

- ▶ Soundness is easy. It requires proving that each rule is valid.

For example:

$$\frac{A \quad B}{A \wedge B} \quad [\wedge I]$$

is valid because $A, B \models A \wedge B$

- ▶ Completeness is harder

We will not prove them here

Conclusion

What did we cover today?

- ▶ semantics of propositional logic
- ▶ satisfiability & validity
- ▶ truth tables
- ▶ soundness & completeness

Further reading

- ▶ Chapter 6 of http://leanprover.github.io/logic_and_proof/

Next time?

- ▶ equivalences
- ▶ normal forms